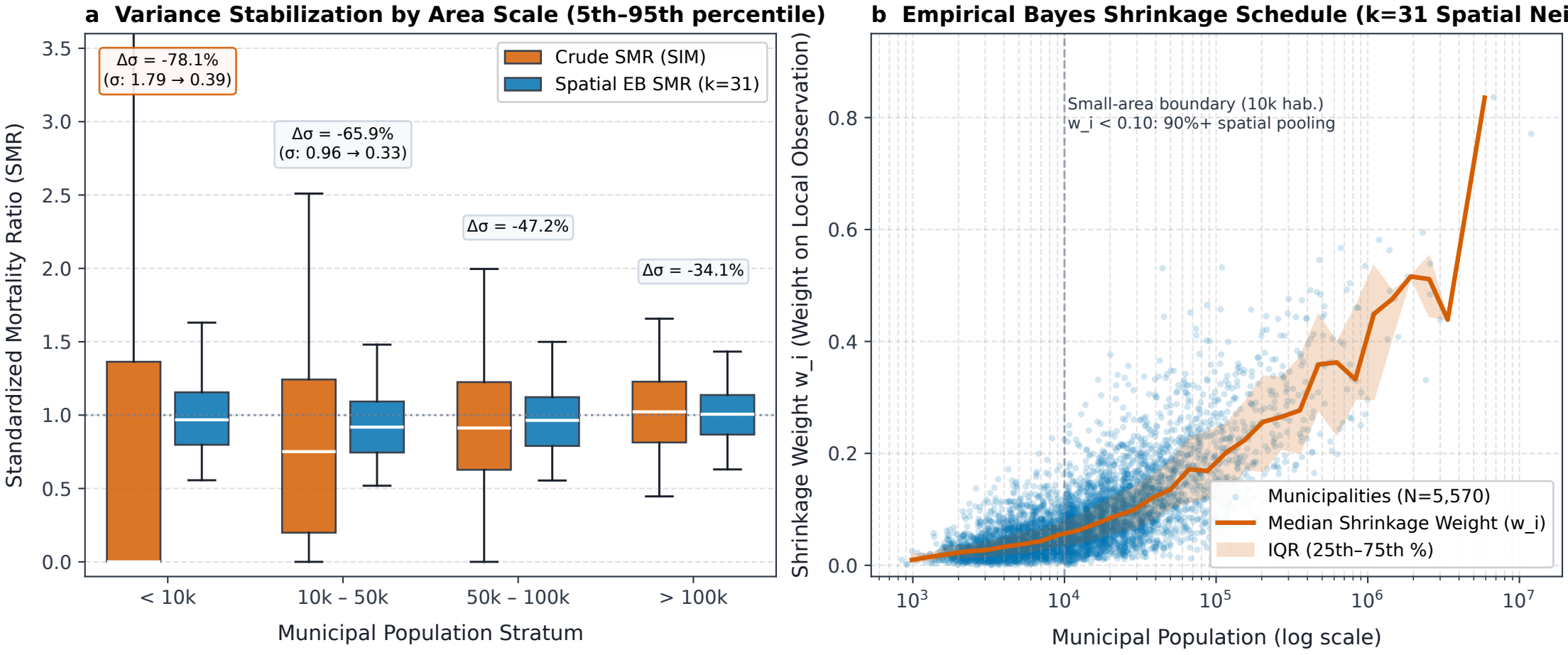

a Variance Stabilization by Area Scale (5th–95th percentile)
Δσ = -78.1%
(σ: 1.79 → 0.39)
Δσ = -65.9%
(σ: 0.96 → 0.33)
Δσ = -47.2%
Δσ = -34.1%
Crude SMR (SIM)
Spatial EB SMR (k=31)
Standardized Mortality Ratio (SMR)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
< 10k
10k – 50k
50k – 100k
> 100k
Municipal Population Stratum
b Empirical Bayes Shrinkage Schedule (k=31 Spatial Nei
Small-area boundary (10k hab.)
w_i < 0.10: 90%+ spatial pooling
Shrinkage Weight w_i (Weight on Local Observation)
0.8
0.6
0.4
0.2
0.0
Municipalities (N=5,570)
Median Shrinkage Weight (w_i)
IQR (25th–75th %)
10^3
10^4
10^5
10^6
10^7
Municipal Population (log scale)